Integrale curbilinii de speța a II-a

• Să se calculeze $\int_\gamma \alpha$ (integrale curbilinii de speța a II-a)

19. $\alpha = (x^2+y^2)\,dx + (x^2-y^2)\,dy$

$\gamma(t): \begin{cases} x(t) = \sqrt{t} \\ y(t) = \sqrt{t+1} \end{cases}$, $t \in [1,4]$

$\gamma'(t): \begin{cases} x'(t) = \frac{1}{2\sqrt{t}} \\ y'(t) = \frac{1}{2\sqrt{t+1}} \end{cases}$

$P(x,y) = x^2+y^2 \Rightarrow P(\gamma(t)) = 2t+1$

$Q(x,y) = x^2-y^2 \Rightarrow Q(\gamma(t)) = -1$

$$\int_\gamma \alpha = \int_\gamma (P\,dx + Q\,dy) = \int_1^4 (P(\gamma(t))\cdot x'(t) + Q(\gamma(t))\cdot y'(t))\,dt = \int_1^4 \left(t^{\frac{1}{2}} + \frac{1}{2\sqrt{t}} - \frac{1}{2\sqrt{t+1}}\right) dt = \frac{17}{3} - \sqrt{5} + \sqrt{2}$$

20. $\alpha = \frac{1}{1+y^2}\,dx + \frac{y}{1+x^2}\,dy$

$\gamma(t): \begin{cases} x(t) = t^2 \\ y(t) = t \end{cases}$, $t \in [0,1]$

$\gamma'(t): \begin{cases} x'(t) = 2t \\ y'(t) = 1 \end{cases}$

$P(x,y) = \frac{1}{1+y^2} \Rightarrow P(\gamma(t)) = \frac{1}{1+t^2}$

$Q(x,y) = \frac{y}{1+x^2} \Rightarrow Q(\gamma(t)) = \frac{t}{1+t^4}$

$$\int_\gamma \alpha = \int_\gamma (P\,dx + Q\,dy) = \int_0^1 (P(\gamma(t))\cdot x'(t) + Q(\gamma(t))\cdot y'(t))\,dt = \int_0^1 \frac{2t}{1+t^2}\,dt + \frac{1}{2}\int_0^1 \frac{2t}{1+t^4}\,dt =$$

$$= \ln(1+t^2)\Big|_0^1 + \frac{1}{2}\operatorname{arctg}(t^2)\Big|_0^1 = \ln 2 + \frac{\pi}{8}$$

21. $\alpha = \sqrt{x}\,dx + xy^2\,dy$

$\gamma: y = x^2$, $x \in [0,1]$

$\gamma(t): \begin{cases} x(t) = t \\ y(t) = t^2 \end{cases}$, $t \in [0,1]$

$\gamma'(t): \begin{cases} x'(t) = 1 \\ y'(t) = 2t \end{cases}$, $t \in [0,1]$

$P(x,y) = \sqrt{x} \Rightarrow P(\gamma(t)) = \sqrt{t}$

$Q(x,y) = xy^2 \Rightarrow Q(\gamma(t)) = t^5$

$$\int_\gamma \alpha = \int_\gamma (P\,dx + Q\,dy) = \int_0^1 (P(\gamma(t))\cdot x'(t) + Q(\gamma(t))\cdot y'(t))\,dt = \int_0^1 (t^{\frac{1}{2}} + 2t^6)\,dt = \frac{2}{3} + \frac{2}{7} = \frac{20}{21}$$

22. $\alpha = (1-y)\,dx + (x-1)\,dy$

$\gamma: x^2 + y^2 - 2x = 3 \iff \gamma: x^2 - 2x + 1 + y^2 = 4 \iff \gamma: (x-1)^2 + y^2 = 2$

$$r(t): \begin{cases} x(t) = 1 + 2\cos t \\ y(t) = 2\sin t \end{cases}, \quad t \in [0, 2\pi]$$

$$r'(t): \begin{cases} x'(t) = -2\sin t \\ y'(t) = 2\cos t \end{cases}$$

$P(x,y) = 1 - y \Rightarrow P(r(t)) = 1 - 2\sin t$

$Q(x,y) = x - 1 \Rightarrow Q(r(t)) = 2\cos t$

$$\int_\gamma \alpha = \int_\gamma (P\,dx + Q\,dy) = \int_0^{2\pi} (P(r(t))\cdot x'(t) + Q(r(t))\cdot y'(t))\,dt = \int_0^{2\pi} (-2\sin t + \underbrace{4\sin^2 t + 4\cos^2 t}_{4})\,dt = 8\pi$$

23. $\alpha = (4x - 6y)\,dx + (6x + 9y)\,dy$

$\gamma: \frac{x^2}{9} + \frac{y^2}{4} = 1, \quad x \le 0$

$$r(t): \begin{cases} x(t) = 3\cos t \\ y(t) = 2\sin t \end{cases}, \quad t \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$$

$$r'(t): \begin{cases} x'(t) = -3\sin t \\ y'(t) = 2\cos t \end{cases}, \quad t \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$$

$P(x,y) = 4x - 6y \Rightarrow P(r(t)) = 12(\cos t - \sin t)$

$Q(x,y) = 6x + 9y \Rightarrow Q(r(t)) = 18(\cos t - \sin t)$

$$\int_\gamma \alpha = \int_\gamma (P\,dx + Q\,dy) = \int_{\pi/2}^{3\pi/2} (P(r(t))\cdot x'(t) + Q(r(t))\cdot y'(t))\,dt = 36\int_{\pi/2}^{3\pi/2} (1 - \sin 2t)\,dt = 36\pi$$

24. $\alpha = (3x^2 + y\cos x)\,dx + (\sin x + 4y^3)\,dy$, $A(0,2,3)$

1 - forma diferențială α este exactă pe $\mathbb{R}^2$ dacă există o funcție f (potențial scalar) astfel încât:

$$\begin{cases} \frac{\partial f}{\partial x} = 3x^2 + y\cos x \\ \frac{\partial f}{\partial y} = \sin x + 4y^3 \end{cases} \Big| \int dx \Rightarrow f(x,y) = x^3 + y\sin x + c(y) \Rightarrow \frac{\partial f}{\partial y} = \sin x + c'(y) \Big| \Rightarrow c'(y) = 4y^3 \Rightarrow c(y) = y^4 + K, K \in \mathbb{R}$$

$\Rightarrow f(x,y) = x^3 + y\sin x + y^4 + K, \quad K \in \mathbb{R}$

f trece prin $A(0,2,3) \iff f(0,2) = 3 \iff 16 + K = 3 \iff K = -13 \Rightarrow f(x,y) = x^3 + y\sin x + y^4 - 13$

25. $\vec{V}' = (\sin(xy) + xy\cos(xy) + 2x,\ x^2\cos(xy) + 2y)$

$\gamma(t): \begin{cases} x(t) = t\cos t \\ y(t) = 2\sin t + \cos^3 t \end{cases}, t \in [0, \frac{\pi}{2}]$

γ curbă deschisă, netedă, simplă

Verifică dacă 1-f.d. asociată câmpului vectorial este exactă, adică dacă există o funcție f (potențial scalar) astfel încât:

$$\begin{cases} \frac{\partial f}{\partial x} = \sin(xy) + xy\cos(xy) + 2x \\ \frac{\partial f}{\partial y} = x^2\cos(xy) + 2y \end{cases} \Big| \int dy \Rightarrow f(x,y) = x\sin(xy) + y^2 + c(x) \Rightarrow \frac{\partial f}{\partial x} = \sin(xy) + xy\cos(xy) + c'(x) \Big| \Rightarrow$$

$\Rightarrow c'(x) = 2x \Rightarrow c(x) = x^2 + K,\ K \in \mathbb{R} \Rightarrow f(x,y) = x\sin(xy) + y^2 + x^2 + K,\ K \in \mathbb{R} \Rightarrow$ 1-f.d. este exactă

$\Rightarrow$ Pot aplica independența de drum

$$\int_\gamma df = f(\gamma(\tfrac{\pi}{2})) - f(\gamma(0)) = f(0,2) - f(0,1) = 4 + K - 1 - K = 3 = \int_\gamma \vec{V}'\,d\vec{r}$$

26. $\vec{F}' = (x, -\sin y)$

$\gamma = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 - 2x = 0,\ y \le x\}$, orientată în sens trigonometric

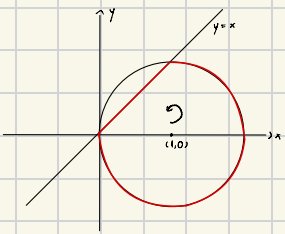
Verifică dacă 1-f.d. asociată câmpului de forțe este exactă

$$\begin{cases} \frac{\partial f}{\partial x} = x \\ \frac{\partial f}{\partial y} = -\sin y \end{cases} \Big| \int dx \Rightarrow f(x,y) = \frac{x^2}{2} + c(y) \Rightarrow \frac{\partial f}{\partial y} = c'(y) \Big| \Rightarrow c'(y) = -\sin y \Rightarrow c(y) = \cos y + K,\ K \in \mathbb{R} \Rightarrow$$

$\Rightarrow f(x,y) = \frac{x^2}{2} + \cos y + K,\ K \in \mathbb{R} \Rightarrow$ 1-f.d. e exactă ①

$x^2 + y^2 - 2x = 0 \Rightarrow (x-1)^2 + y^2 = 1$

$\gamma(t): \begin{cases} x(t) = 1 + \cos t \\ y(t) = \sin t \end{cases}, t \in [-\pi, \frac{\pi}{2}]$



① $\Rightarrow \int_\gamma df = f(\gamma(\frac{\pi}{2})) - f(\gamma(-\pi)) = f(1,1) - f(0,0) = \cos 1 - \frac{1}{2} = \int_\gamma \vec{F}'\,d\vec{r}$